

JSON-Based Sequencer and Customizable Assets Specification Proposal

Overview

This specification defines how the `sequence.json` drives the immersive experience, managing localized audio, visuals, and text instructions, while keeping static UI assets (like the panel background and welcome screen graphics) in a separate, clearly structured folder for allowing the customization of the visual appearance of the app.

The **Sequencer**, as defined by the `sequence.json` file, must be capable of the following:

- Sending **Arduino commands**
- Displaying and hiding **images or videos** (one at a time only, with preferred resolution)
- Showing/hiding **UI panels and instructional text**
- Playing localized **audio** in **4 languages** (de, en, fr, it)

On the other hand, the following **Static assets** can be defined :

- Panel **background** image
- **Start button** image (on/off states)
- **Language buttons** images
- The **font** used in UI panels and its **color**

Sequence Step Format

Each timeline event is represented as a step in the JSON. The format for each step is as follows:

```
{  
  "time": 0.0,
```

```
"textKey": "Welcome",  
"audio": "welcome.mp3",  
"visual": "instructions.png"  
"actions": []  
}
```

Field Descriptions:

- **time**: Time (in seconds) since the experience starts (triggered when both users are ready).
- **textKey**: A key that maps to a localized instruction string. Also shows the text panel if not already showing.
- **audio**: File name of the audio (.ogg) instruction. Separate files must exist for each language in their respective folders
- **visual**: Image (.png) or video (.mp4) — Loads the visual, use `ShowVisual` to show it
- **actions**: List of actions to perform at this step.

Supported Actions

You can include one or more of the following actions in any step:

- `ShowVisual`
- `HideVisual`
- `HidePanel`
- `WallOn`
- `WallOff`
- `MirrorOn`
- `MirrorOff`
- `StartExperience`
- `EndExperience`

Language Support

- Language selection uses country codes: `en`, `de`, `fr`, `it`, etc.
- Audio is stored in `/Sequence/Audio/{lang}/`
- Translation JSONs in `/Sequence/Translations/{lang}.json`

- Flags in `/Static/` using `flag_{code}.png`

Folder Structure

Make sure the following folder structure is followed to ensure all the assets are loaded correctly :

```
/TheMachineToBeAnother/  
├─ TheMachineToBeAnother.exe  
├─ /Content/  
│   ├─ /Sequence/  
│   │   └─ sequence.json  
│   │   └─ Audio/  
│   │       └─ en/  
│   │           └─ welcome.ogg  
│   │       └─ de/  
│   │           └─ welcome.ogg  
│   │       └─ fr/  
│   │           └─ welcome.ogg  
│   │       └─ it/  
│   │           └─ welcome.ogg  
│   │   └─ Images/  
│   │       └─ instructions.png  
│   │   └─ Videos/  
│   │       └─ instructions.mp4  
│   └─ Translations/  
│       └─ en.json  
│       └─ de.json  
│       └─ fr.json  
│       └─ it.json  
└─ /Static/  
    └─ panel_background.png  
    └─ start_button_on.png  
    └─ start_button_off.png  
    └─ flag_en.png  
    └─ flag_de.png  
    └─ flag_fr.png
```

```
|      └─ flag_it.png
|      └─ font_color.txt
```